Developing Shopping and Dining Walking Indices Using POIs and Remote Sensing Data in Rangpur Sadar

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Abstract

The popularity of walking as a mode of transportation is growing steadily. Scholars have devised a slew of indices and ratings to gauge a city's walkability. Walking indices and scores are most often used in urban planning and service provision, regional accessibility, and physical activity. Walking is a more environmentally friendly means of transportation, and governments all over the globe are working to make their cities more walkable. In recent years, walkability has been a study issue because of its relevance. The built environment's impact on walking as a primary mode of transportation has been extensively studied. As a result of a review of previous research, this study devised an indicator-based framework for quantifying walkability. In order to arrive at an overall walkability score, the indications were weighted, normalized, and then combined. Most of the findings, including LST, shopping malls, and restaurants are concentrated in the central part of the region.

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1 Introduction

The most frequent mode of transportation is walking, yet it isn't usually considered a comprehensive means. The structure of the vehicle is referred to as "pedestrian." Vehicle-based models seldom treat pedestrians equally. The handling of transport modes is heavily influenced by solid advocacy organizations, facts regarding the method, its performance and effects, and the persons and groups impacted. Most journeys need some walking, and walking affects access to many amenities. The usage of public transportation necessitates walking both to and from the vehicles. People who utilize private transportation must also walk to and from their vehicles and their destinations. (Wigan, 1995)

Walkability is gaining popularity because of the poor condition of public places for pedestrians in metropolitan areas (footpaths, pavements, etc.). The streets should be open to pedestrians, not only those in automobiles and parking spots, as has been the case in many cities. This includes those who are using public transportation. Pedestrian and bicycle models of individual mobility should be available to all inhabitants of metropolitan areas. Walkability is also linked to health and environmental concerns and economic advantages. There are four critical requirements for a city to be considered "walkable": security, functioning, appeal, and convenience. (Katarzyna TUROŃ, 2017)

It hasn't proven easy to develop a widely acknowledged strategy for evaluating walkability. It is important to note that Abley (2005) defined their scopes such that "reviewing" was largely qualitative and used to study an existing state, while "rating" was accomplished in a quantitative method and used to evaluate an existing or a planned scenario. It is possible to do "auditing" qualitative and quantitatively for both current and future situations. (Hediye TuydesYaman, 2018)

Alternative urban forms, including transit-oriented projects, new urbanist neighborhoods, and walkable communities, are needed to maintain the current mass transportation scenario. The purpose of this research was to examine the walking habits of commuters in terms of their requirements, perceptions, and attitudes toward public transportation. The goal is to reduce the number of individuals who rely only on automobiles to move about. Planners must better grasp walking environmental planning to achieve this goal. (Pawinee Iamtrakul, 2021)

Another area of walkability evaluation is regional accessibility-oriented walkability. Walkability was measured by analyzing the infrastructure and track and the surrounding environment and activities. Urbanity was also taken into consideration. Planners and plannersto-be may utilize this walkability index with ease. Based on community boundaries, subway station density, land use mix, number of street junctions, and the ratio of the retail building floor size to retail land area, Freeman et al. established a neighborhood walkability scale. (Yingbin Deng, 2020)

High residential density, land-use mix, and street connectivity have been utilized to identify built environment variables that may affect commute and leisure-time physical activity in conjunction with each other. Walkability and physical activity are linked in high-income nations, according to several studies. (Rodrigo Siqueira, 2013) Density, variety, and accessibility are the main forces behind PA. People who live in areas with more significant housing densities and more diverse land uses are more likely to walk to their destinations. (Yehua Dennis Wei, 2016)

The health, economy, and overall livability of a region are reflected in its walkability. It's not only about how far you can walk but also about how safe, convenient, secure, visually appealing, and well-connected the neighborhood is. Separated land use, dead-end streets, and poorly planned development layouts are the wrong places to stroll. A walkable environment has flat sidewalks, safe junctions, narrow roadways, suitable disposal facilities, and sufficient lighting, as well as a lack of barriers in the path of pedestrian traffic. There must be a local culture and a built-in environment restricting more unfettered behavior. (PRAGIA MINHAS, 2017)

We wanted to create shopping- and dining-specific walking indices (SWI and DWI) that considered both the pedestrian's intention (to shop or dine out) and their immediate physical surroundings (such as vegetation, water, or a range of temperatures). (Yingbin Deng, 2020)

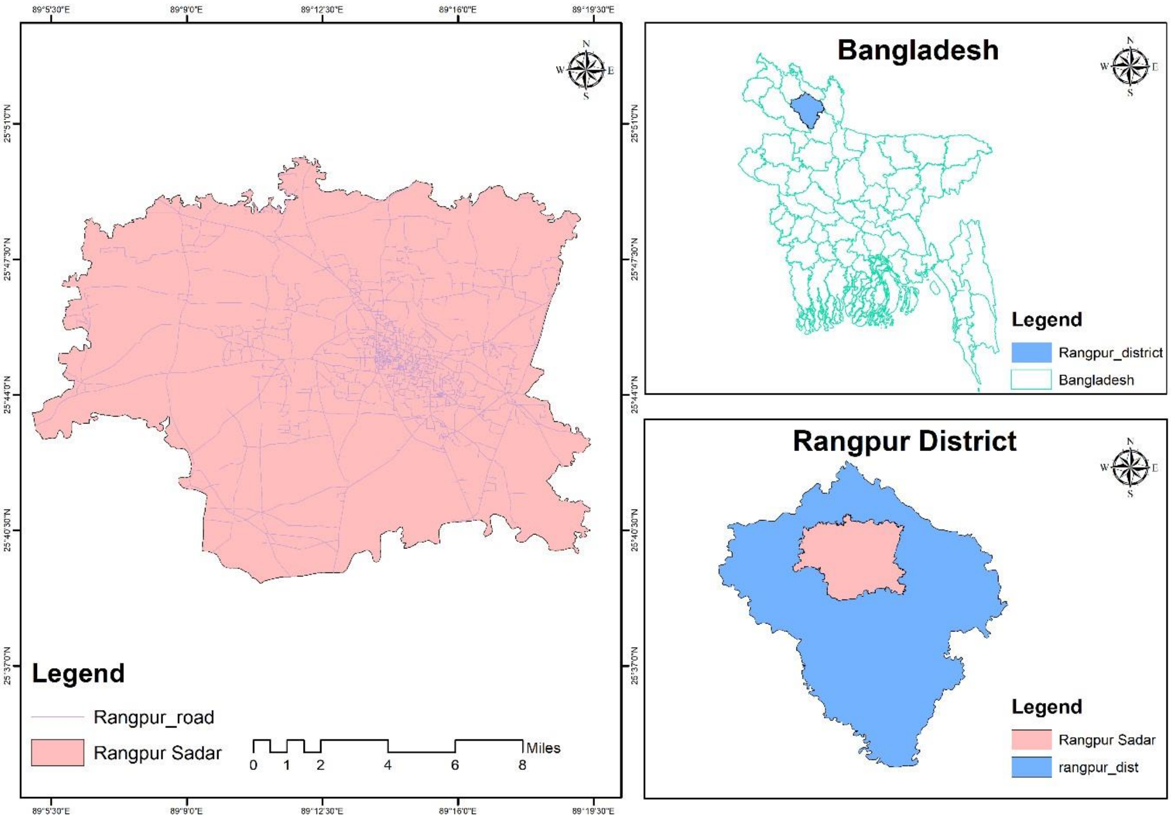
2 Materials and Methods

2.1 Study Area

Rangpur Division became Bangladesh's seventh division on January 25, 2010. Before that, it had been the northern eight districts of the Rajshahi Division. The Rangpur division consists of eight districts, that is Rangpur, Dinajpur, Kurigram, Gaibandha, Nilphamari, Panchagarh, Thakurgaon, and Lalmonirhat. There are 58 Upazillas or subdistrics under these eight districts. Rangpur is the northernmost division of Bangladesh and has a population of 15,665,000 according to 2011 Census. Transportation - this division is well communicated with other division. Road, Rail and Airport communication is available in this division.

The shopping and dining orientated street indices were built using POI data and remotely sensed data. In 2020, we collected POI data from my map, which included shopping malls, retail stores, and other types of restaurants. It has a programming interface for applications. Developers can use the (API) to get the POI data. Employing coordinate information to transform them to georeferenced vector data A scene from the film On November 15, 2020. The normalized difference vegetation index (NDVI) and the normalized difference water index (NDWI)indices (NDWI). The Landsat 8 Thermal Infrared Sensor (TIRS, 30-m spatial resolution) captured this image.

The land surface was retrieved using a cloud cover percentage of collected on November 15, 2020.temperature. In November, Landsat 8 pictures were chosen for two projects. (1) The image acquisition date is approaching. Landsat 8 are separated by only one day. (2) The picture quality. During the spring and summer, there are limited high-quality Landsat 8 photos available. Summers are hot and humid, with a lot of gloomy days. Vector data from the streets and bus/subway data. The data for the stations came from my map. Downloaded street view images. The walking environment of each street segment was assessed using data from Tencent map. Data in its entirety in the zone 46N, they were reprojected to the Universal Transverse Mercator Projection (UTM).



Map 01: Study Area

2.2 Index Construction

In this work, we used remote sensing data and POI data to create retail and dining walking indices using six steps: Processing of street data. Index construction was based on street data. Processing of POI data. The nearest street segment was linked to the POI data, and the POI density (restaurants) was then calculated. density and shopping store POI density of each street segment were estimated, and the results were compared. Third, the NDVI in addition to the use of NDWI processing. Landsat 8 imagery was used to calculate NDVI and NDWI. In order to discriminate between vegetation and water, the NDVI and NDWI thresholds were defined. The density of vegetation Each street segments and water density were determined. The temperature of the Earth's surface with regard to thermal infrared images from Landsat 8 were used to reverse the temperature of the Earth's surface. (Yingbin Deng, 2020)

In this case, the term "sensor" (TIRS). Then, a street segment's average ground surface temperature was computed. Take a picture of cells in a specific area. In the end, the results were analyzed using a categorization scheme. Thermometer readings. Distance calculation between bus and metro stops. Using the linear distance function. Calculation of SWI and DWI.

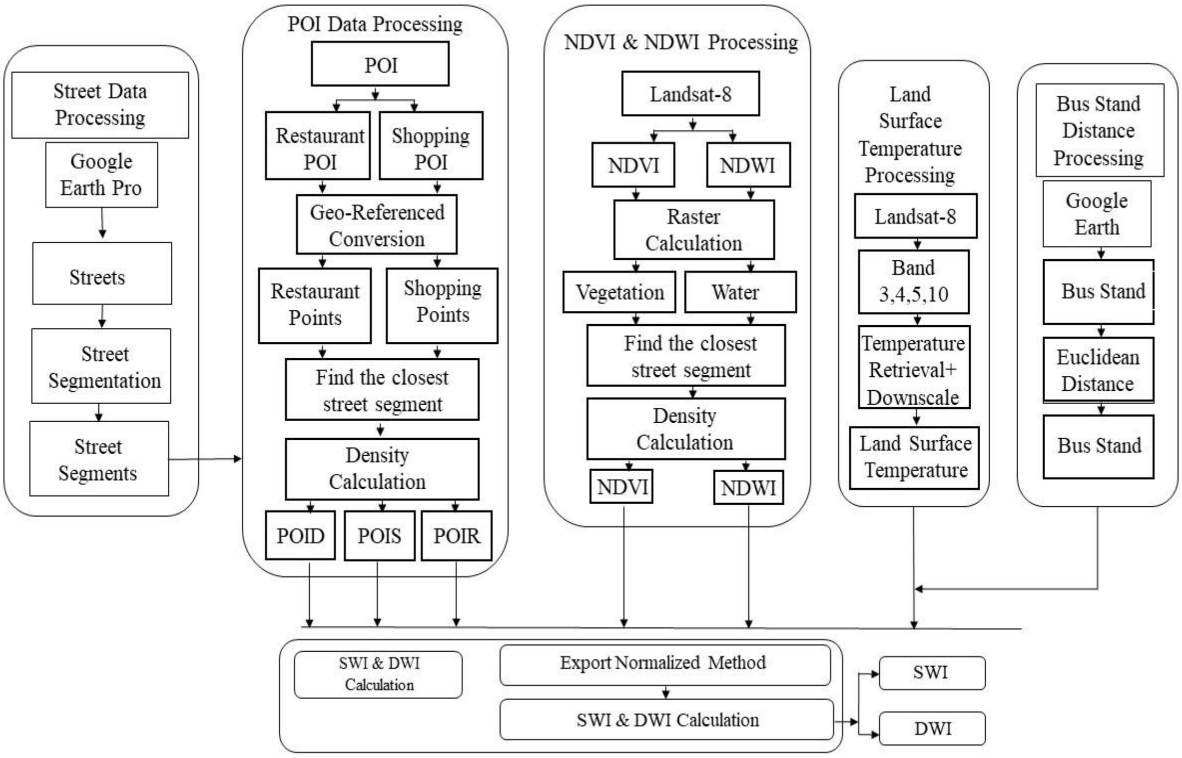


Figure 1. Flow chart of walking index construction

2.2.1 Street Data Processing

Before creating the index, this study filtered the street vector data. We removed all of the major thoroughfares that are inaccessible to people walking. Pedestrian-friendly roadways, such as footpaths, residential streets, side streets, and service roads, were preserved. The purpose of a test This review was limited to providing generic information for streets longer than one kilometre. (Yingbin Deng, 2020)

Take, for instance, a 300-meter stretch of road: Vegetation has grown across half of the street. The first portion of the road is ideal for pedestrians and gives them access to POIs. the calculated indexes of walking. Only a single figure (presumably the average of the entire environment) was available in earlier study.it is difficult to precisely assess the favourable atmosphere created by this long roadway, and as a result, The unfavourable environment of the other side of the roadway (which has a significant concentration of plants and POIs) a second half (few vegetation or POIs).

2.2.2 POI Data Processing

Two types of POI data were used in this study: dining and shopping POIs. There are numerous sub-types of POI. POI data were geocoded from a text file utilizing their X and Y coordinate information. A point shapefile containing UTM projection information was then created from these data.

For each POI point, the Near function in ArcMap (ArcGIS) software (Environmental Systems Research Institute, ESRI) was used to find the nearest street. After that, we tallied up the number of POIs found along each street segment. Finally, the density of POIs for dining and shopping centres (POIR and POIS) in order to arrive at the following results, we used these Equations.

$$POIR = (\text{Number of Restaurant POI})/(\text{Street Segment Length}) \tag{1}$$

$$POIS = (\text{Number of Shopping store POI})/(\text{Street Segment Length}) \tag{2}$$

2.2.3 NDVI and NDWI Processing

The NDVI and NDWI were derived from a multispectral image captured by the GaoFen-2 satellite. It was found that the NIR and RED bands were used to calculate NDVI, whereas the NIR band and GREEN band were used to calculate NDWI. In Equations (3) and (4), the derivation is shown:

$$NDVI = (NIR - RED)/ (NIR + RED) \tag{3}$$

$$NDWI = (GREEN - NIR)/ (GREEN + NIR) \tag{4}$$

where NDVI and NDWI are the normalized difference vegetation index and normalized difference water index, respectively. NIR, RED, and GREEN are the near-infrared band, red band, and green band in the GaoFen-2 multispectral image.

2.2.4 Land Surface Temperature Processing

Land Surface Temperature (LST) is the temperature of the earth's surface which is the radiative skin temperature of the land derived from infrared radiation. The "skin" temperature is the top surface temperature in the bare soil condition. It is also effective in emitting vegetation canopy temperature.

A single channel approach using the Landsat 8 Thermal Infrared Sensor (TIRS) and Band 11 of the atmospheric correction method radiative transfer equation reversed the original land surface temperature. Then, the temperature on the ground was Kriging model and geostatistics were used to reduce the spatial resolution of the data to a 10-meter grid. NDVI, normalized difference building, impervious surface fraction, vegetation fraction, soil fraction, and so on are all terms used to describe these many variables. Digital elevation model (DEM), index (NDBI), density (NDWI) and building density. The impenetrable ones the spectral mixture model for Landsat 8 Operational Land can be used to compute the surface fraction. Multispectral photos from the Imager (OLI, 30-m spatial resolution) This is very much like the preparation of vegetation by deleting any temperature cells that were farther away from the nearest streets than 15 meters. The rest of the Values associated with each point in the vector were assigned to the temperature cells that were turned into vector points. The mean Using the remaining data points, the land surface temperature of the relevant area was calculated for street sections. (Yingbin Deng, 2020)

$$TOA\ (L) = ML \times Q(cal) + AL - O \quad (5)$$

$$BT = \frac{K_2}{\ln\left(\frac{K_1}{TOA} + 1\right)} - 273.15 \quad (6)$$

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (7)$$

$$PV = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}}\right)^2 \quad (8)$$

$$\varepsilon = 0.004 \times P_v + 0.986 \quad (9)$$

$$LST = BT / (1 + \lambda * BT / c2 * \ln(\varepsilon)) \quad (10)$$

2.2.5 Bus Station Distance Processing

Because the streets were divided into 30-m segments and some non-walking parts were omitted, the distance between bus terminals and each street segment must also be considered. It was difficult to compute the network distance because the street's continuity had been broken.

Therefore, we estimated the distance between the bus station and the street section using the linear line distance. The time it would take to walk 1500 meters at the average walking speed of 120 meters per minute is simply untenable while trying to locate a business or restaurant following a trip on the bus or train. As a result, the sections of streets that are within 1500 meters of a bus station received a positive score, while those that are further away received a zero.

2.2.6 SWI and DWI Calculation

After calculating the POI density, vegetation coverage, water coverage, subway/bus station distance (transport score), and temperature, we constructed the shopping and dining walking indices.

The walking index was constructed using five variables, namely, the street segment-based POI density (restaurant density or shopping point density), vegetation density, water density, transport score (distance to bus/subway station), and average temperature. Expert scoring methods were employed to assign weighting scores to different variables. The detailed weighting scores are shown.

Variables	POI	Vegetation	Water	Distance	Temperature
Scores	0.6	0.3	0.1	0.1	–0.1

The POI density variable was assigned the highest weighting score of 0.6. Tree shadows can provide a cooling effect for pedestrians in the summer. Other vegetation types can also beautify the landscape in urban areas. Thus, we set the second-highest weighting score of 0.3 to vegetation density. Water density also has a cooling function and ornamental value, as pedestrians feel comfortable when they walk around a water area. Hence, we set 0.1 as the weighting score for water. The weighting value for bus/subway station distance was set to 0.1 because we assumed that the influence of walking distance was smaller than the impact of the walking environment (vegetation density) and walking purpose (POI density). Extremely hot and cold temperatures make pedestrians feel uncomfortable. Therefore, the weighting score for temperature was negative (0.1) in this study. The resulting shopping walking index (SWI) and dining walking index (DWI) are expressed in Equation and Equation respectively:

$$SWI = POI_S * 0.6 + NDVI * 0.3 + NDWI * 0.1 - Distance*0.1 - Temperature *0.1 + Density* 0.1 \quad (11)$$

$$DWI = POI_D * 0.6 + NDVI * 0.3 + NDWI * 0.2 - Distance*0.1 - Temperature *0.1 + Density* 0.1 \quad (12)$$

Shopping Walking Index	**Recommendation**	**Description**
0 – 0.128	Deprecated	Low POI density or no vegetation and water area, long
0.128 – 0.233	Acceptable	Not many POIs, vegetation and water cover is low
0.233 - 0.35	Recommended	Some POIs, medium density of vegetation or water cover, medium distance to bus/subway station, suitable temperature
0.35 – 0.66	Very recommended	Many POIs and good walking environment. High density of vegetation or water cover, short distance to bus/subway station, suitable temperature
0.66 - 1	Highly Recommended	Many POIs and very good walking environment. High density of vegetation and water cover, short distance to bus/subway station, suitable temperature

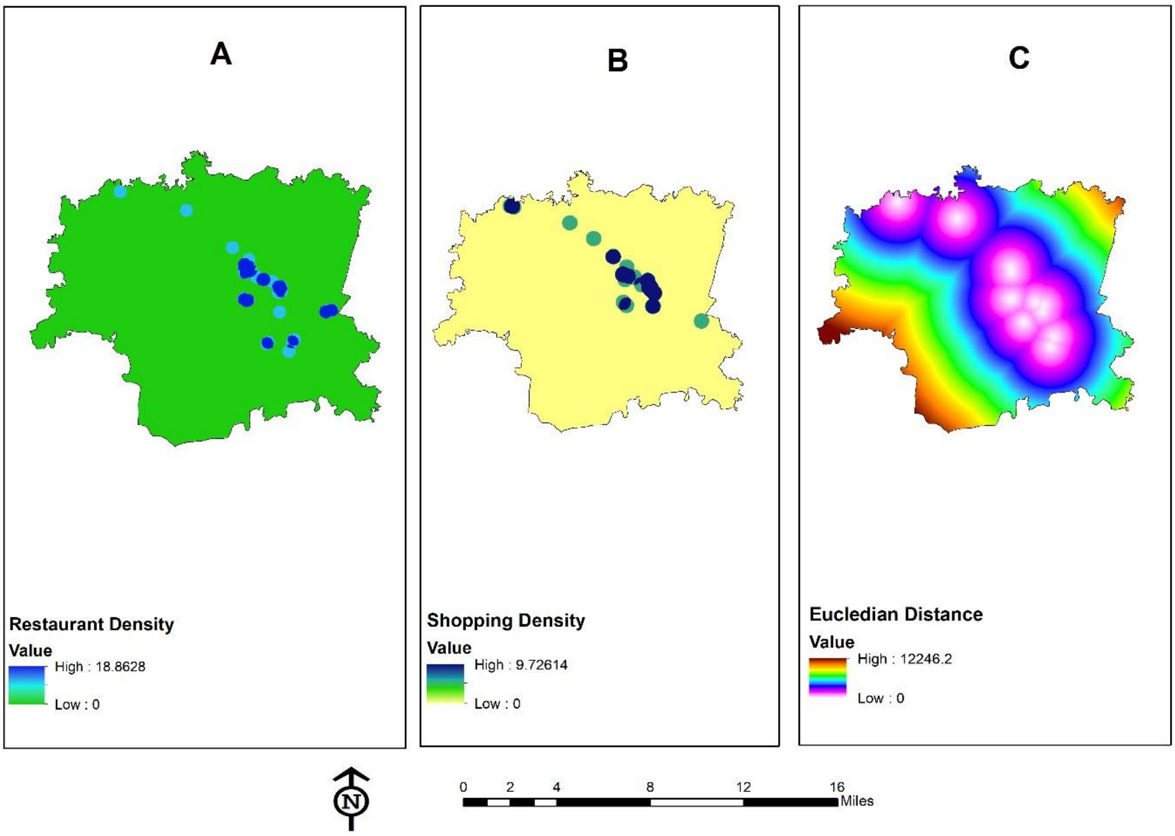
Table:1 Shopping Walking Index Categories

Dining Walking Index	**Recommendation**	**Description**
0 – 0.212	Deprecated	Low POI density or no vegetation and water area, long
0.212 – 0.23	Acceptable	Not many POIs, vegetation and water cover is low

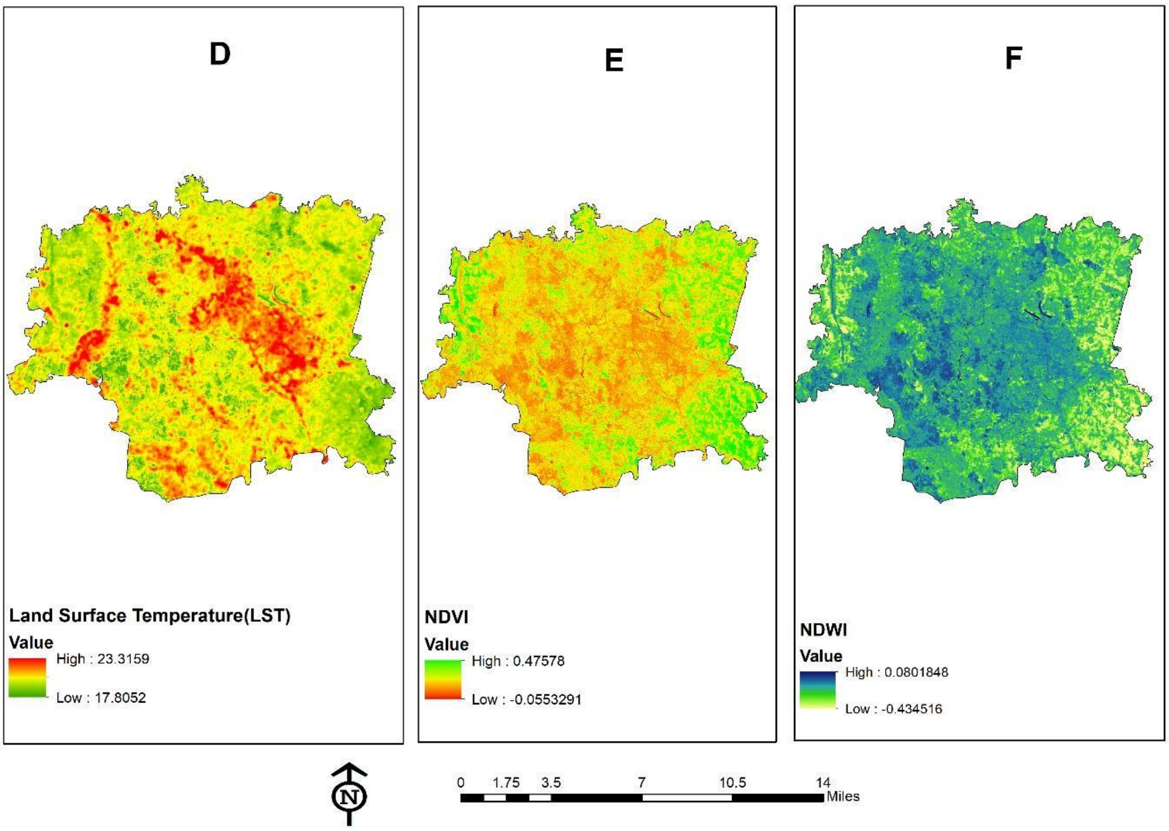
0.23 - 0.26	Recommended	Some POIs, medium density of vegetation or water cover, medium distance to bus/subway station, suitable temperature
0.26 – 0.34	Very recommended	Many POIs and good walking environment. High density of vegetation or water cover, short distance to bus/subway station, suitable temperature
0.34 - 1	Highly Recommended	Many POIs and very good walking environment. High density of vegetation and water cover, short distance to bus/subway station, suitable temperature

Table:2 Dinning Walking Index Categories

3 Result



Map 02: Indicator Map A) restaurant density B) Shopping density C) Bus distance

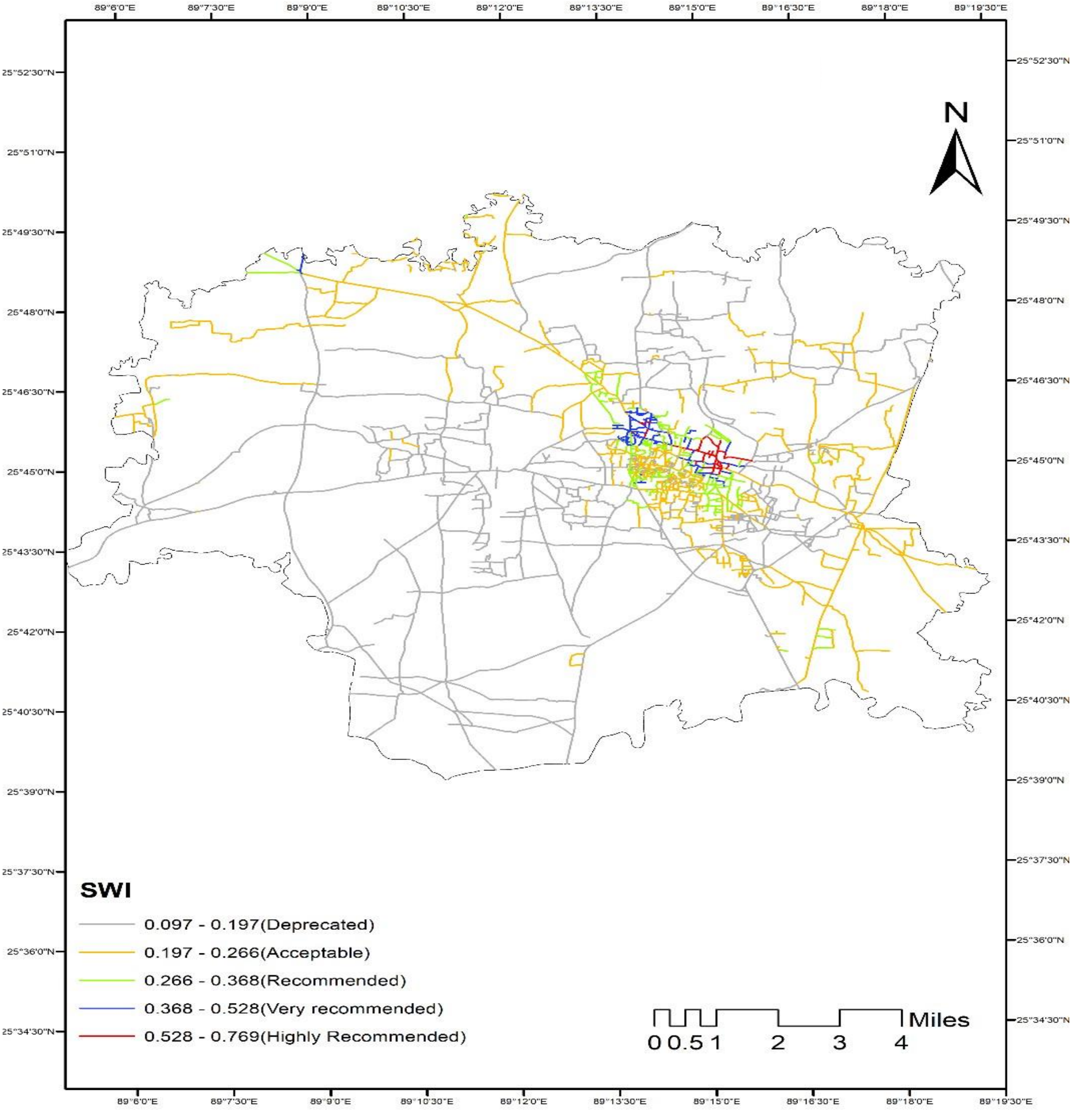


Map 03: Indicator Map D) LST E) NDVI F)NDWI

These above indicator maps were used to calculate the SWI and DWI of the rangpur sadar. Here we can see the shopping density, dinning density and distance from bus stand and NDVI, NDWI and LST map. Here the later index calculated for analysis the willingness of walk in a pedestrian walking environment. From above index NDVI and NDWI is positively correlated with walking where LST and Bus stop distance is negatively correlated with walking. Then we normalize mean value represent the walkable condition on a segment of road.

3.1 Shopping Walking Index (SWI)

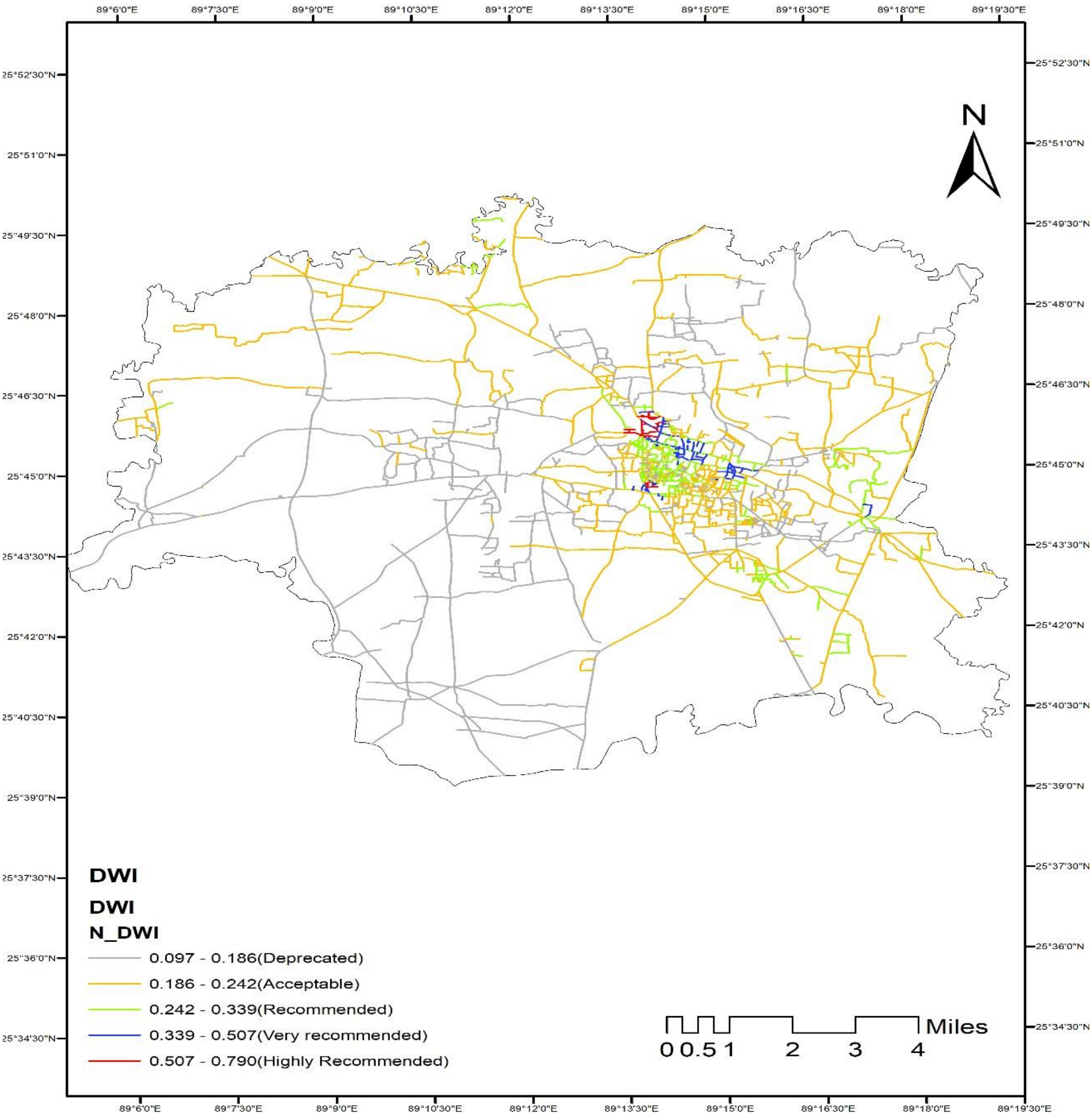
The SWI is a pedestrian-friendly shopping index. Many of the study area's most highly rated street segments are absent. There are more recommended and very recommended street segments than there are recommended and extremely recommended street segments. Outside of the previously mentioned major economic centres, several areas of the country are severely lacking in open space. Most the area's street segments are in deprecated condition.



Map 4: Shopping Walking Index Map

3.2 Dining Walking index (DWI)

The Dining Walking Index (DWI) measures how difficult it is to get from one restaurant to another based on how many there are (Map:05). High DWI values (DWI > 0.5) are representing highly recommended street segment. DWI values are much more common. "Highly recommended" status can be given to a small number of roadway segments. Most of the Recommended and Very Recommended Street segments can be found in downtown areas and other high-traffic areas. Parks and residential areas tend to have DWI values below 0.3



Map 5: Dinning Walking Index Map

3.3 Discussion

Shopping- and dining-oriented street settings were assessed using the Shopping Walking Index (SWI) and the Dining Walking Index (DWI). The transparent and understandable approach to indicator selection, quantification, weighting and index development could also help decision-makers in finding characteristics of the built environment that need improvement to reach high walkability levels (Reisi, Nadoushan, & Aye, 2019). Walking conditions (such as vegetation and water cover), bus/subway station distances, and land surface temperature were taken into consideration when constructing the SWI and DWI. The SWI focuses on shopping, while the DWI centres on restaurants.

The POI data collected from My maps are the primary source used to evaluate the distribution of urban facilities. Since the SWI and DWI are shopping- and dining-oriented, respectively, their POIs are the most important variable for constructing the indices. Unlike the walking score, our proposed walking indices only considered the shopping and dining POIs. In order to encourage people to live in walkable communities, the walking score was created, also uses these POIs to calculate its index.

Vegetation and water cover, two additional variables that may affect the walking environment, are also included in this study. Pedestrians may be more inclined to walk for shopping or dining if they have a cool place to rest and relax. NDVI and NDWI, two metrics derived from high resolution remote sensing pictures, are quick and easy ways to assess vegetation and water cover. Despite their simplicity, both approaches are able to offer equivalent vegetation water cover estimates using multispectral high-resolution pictures

The distance to the bus/subway station is another key element for SWI and DWI construction. street segments that are very close to bus/subway stations will have a score of 1, and segments that are 1500 m away from a bus/subway station will have a score of 0. The connectivity of the road provides great convenience for residents, but this may not hold true for consumers or travellers who aim to find a place for shopping or dinning. Thus, a network-based distance function is not employed in our study

The land surface temperature is also a factor that is negatively correlated with the walking and dinning. People tend to walk more in a cool environment

The density of road intersection is another factor. People walk more in the intersections that is positively correlated.

Two walkability index was defined and developed by assigning weights to the indicators by their importance, and integrating them into a single WI in this study (Reisi, Nadoushan, & Aye, 2019) one for shopping, and the other for dining. However, there are still certain issues that will need to be addressed in future research. In order to provide more particular information, the sort of shopping and/or dining should be described in more detail. For example, different sorts of restaurants and shops can be divided into different categories. For these two metrics, assigning varying weights to different types of shopping and dining could yield more exact results. The effect of built environment intervention on walkability of an area is widely accepted (Reisi, Nadoushan, & Aye, 2019) .The mechanism for calculating the distance travelled by bus or subway must be improved in the future as well. A more precise estimation of the walking distance from the bus or metro stop to the final destination can be obtained using a network-based distance computation. This should be taken into mind in the future as well, since the slope might affect how quickly and comfortably people walk.

The following outcome are generated after calculating shopping walking index (SWI) and dining walking index (DWI) on Rangpur sadar. These are-

- The CBD holds most of the shopping centre and dinning places that promotes highly recommended road segment for walking
- It has also shown from the map that, some part of the road separated from central area also promote recommended and highly recommended road for dinning that makes the area larger for dinning and have a commercial value
- The analysis also shows that the area that promotes walkability are the cities most crowded place that centralized the city in a single unit. Other parts of the cities often neglected and thus cannot contribute to the economy.
- The Walkability index can be useful is in the stage of urban planning. High value of Walkability index means that a particular arrangement of the city supports physical activity of people. The low value of Walkability index means that people very often used cars in the everyday life which results into minimum physical activity (Bhadra et al., 2016).

Despite the applicability of the obtained results for transport planning, some challenges are associated with the investigation, including-

Selection of walkability indicators: Selecting a set of indicators which provides a comprehensive overview of the considered system is challenging (Reisi, Nadoushan, & Aye, 2019).

Weighting: Assigned weights using pairwise comparisons could be highly subjective, as they are subject to experts' judgments

Single index development: it is believed that no single index can answer all questions and there is a need for multiple indicators (Jollands, 2003).

4 Conclusion

Walking indices for retail and dining contexts have been proposed in this work, which includes a shopping walking index (SWI), and a dining walking index (DWI). With the help of relevant POIs (e.g., retail and restaurants), vegetation, water area, bus/subway station distance and land temperature, we were able to create the SWI and DWI. Each POI received a different score based on the expert's opinion, not only to underline its importance, but also to emphasize the street's walkability. Each of the seven levels of the SWI and DWI was created to assist people visualize their options when going shopping or dining.

Though there were some limitations during the study procedure, the results show the level of walkability of rangpur sadar. In developed countries various studies have been done to identify level of walkability for different places, but for developing countries like Bangladesh, it is unprecedented. This is the first approach to develop a model by which walkability can be measured for Rangpur City. Further researches incorporating more precise data and variables could lead to more sophisticate and accurate walkability index model.

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